USN | | | | | | | | | | [22CSE557]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER - 2024

MICROPROCESSORS AND MICROCONTROLLERS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a**	Define the term accumulator in the 8086 microprocessor.	L1	CO1	**02**
	b	Explain the difference between minimum mode and maximum mode in 8086.	L2	CO1	**02**
	c	List the addressing modes of the 8051 microcontroller.	L1	CO3	**02**
	d	Illustrate the purpose of serial communication in the 8051.	L2	CO5	**02**
	e	What is the significance of interrupts in the 8051 microcontroller?	L1	CO4	**02**
		PART-B			**6 X 5=30**
2.	**a**	Write an assembly language program to add two 16-bit numbers in 8086.	L3	CO2	**05**
		OR			
	b	Analyze the instruction set of the 8086 microprocessor and its uses.	L4	CO2	**05**
3.	**a**	Explain the internal architecture of the 8051 microcontroller.	L2	CO3	**05**
		OR			
	b	Write a program in assembly language to move a block of data in 8051.	L3	CO3	**05**
4.	**a**	Evaluate the operation of Timer 0 in the 8051 microcontroller.	L5	CO4	**05**
		OR			
	b	Differentiate between various timer modes in the 8051 microcontroller.	L4	CO4	**05**
5.	**a**	Write a C program to transmit data serially using the 8051 microcontroller.	L6	CO5	**05**
		OR			
	b	Compare UART and RS232 communication protocols in the context of 8051.	L4	CO5	**05**
6.	**a**	Design a program to interface 8051 with a stepper motor and rotate it.	L6	CO6	**05**
		OR			
	b	Explain how the 8051 microcontroller can be used to interface with a 7-segment display.	L4	CO6	**05**
7.	**a**	Write an 8051 program to interface a keypad and display the input on an LCD.	L4	CO6	**05**
		OR			
	b	Explain how RS232 protocol is used in 8051 serial communication.	L2	CO4	**05**

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		PART-C			6 X 10=60
8.	**a**	Describe the architecture of the 8086 microprocessor with a labeled diagram.	L2	CO1	**10**
		OR			
	b	Evaluate the physical memory organization of the 8086 microprocessor.	L5	CO1	**10**
9.	**a**	Write a program in assembly language to convert hexadecimal to decimal using 8086.	L6	CO2	**10**
		OR			
	b	Analyze the role of interrupts in the 8086 microprocessor.	L4	CO2	**10**
10.	**a**	Describe the working of 8051 timers and counters and how they can be used to generate delays.	L2	CO4	**10**
		OR			
	b	Write an assembly language program to generate a square wave using Timer 1 in 8051.	L3	CO4	**10**
11.	**a**	Discuss the process of serial communication in the 8051 microcontroller.	L2	CO5	**10**
		OR			
	b	Evaluate the advantages of using RS232 protocol in 8051 serial communication.	L5	CO5	**10**
12.	**a**	Design a system to interface an ADC with the 8051 microcontroller for data acquisition.	L6	CO6	**10**
		OR			
	b	Write a program in C to interface the 8051 with a keypad and display the input on an LCD.	L3	CO6	**10**
13.	**a**	Write an 8086 assembly program to perform searching and sorting operations.	L3	CO2	**10**
		OR			
	b	Explain the function of the stack in the 8086 microprocessor.	L3	CO2	**10**